

EAT: An Enhancer for Aesthetics-Oriented Transformers

ACMMM 2023 Summary

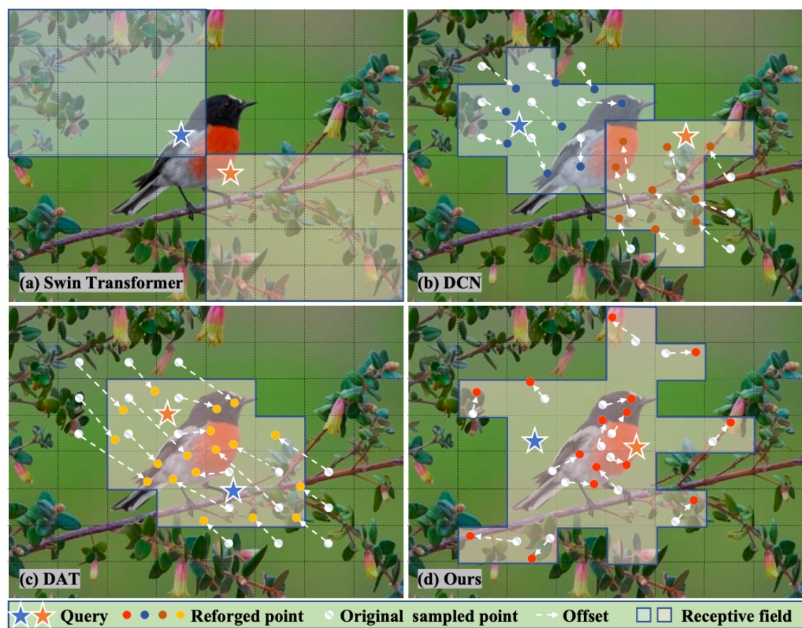
James Hsu – August 8th 2025

Motivation

- Existing Transformer architectures have **not consistently outperformed CNNs** in Image Aesthetic Assessment (IAA) tasks.
- One key reason is the **global, dense self-attention** in Transformers:
 - Tends to **focus heavily on salient foreground regions**.
 - **Ignores subtle aesthetic cues** in background areas.
- This leads to:
 - **High computational cost**
 - **Slow convergence**
 - **Overfitting** on small-scale IAA datasets

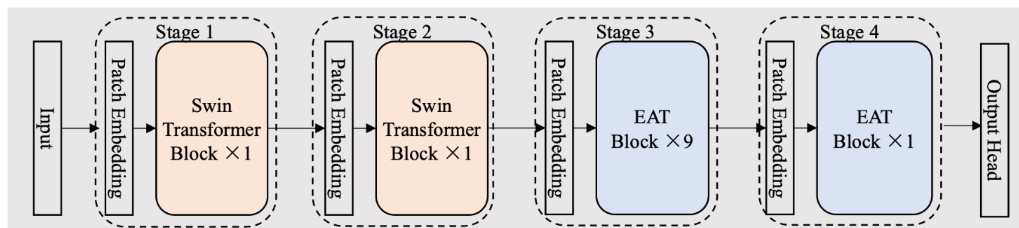
Proposed Method: EAT

- EAT = Enhancer for Aesthetics-Oriented Transformers
- Introduces a **deformable**, **sparse**, and **data-dependent** attention mechanism.
- Learns **offsets** to dynamically select important positions for aesthetic assessment.
- Enables the model to **fine-tune attention distribution** based on image content.

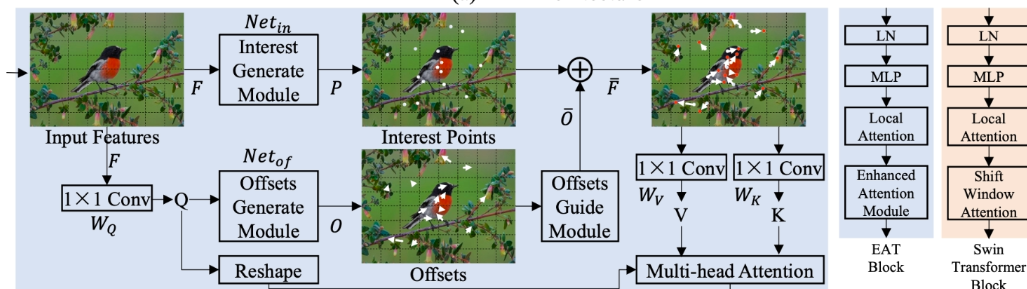


Enhanced Attention Module

- Interest Generation Module
- Offsets Generate Module
- Offsets Guide Module



(a) EAT Architecture



(b) Enhanced Attention Module

Figure 4: A typical structure, denoted as EAT, incorporates the proposed enhancement strategy. (a) Illustrates the overall architecture of EAT and (b) shows feature processing for the enhanced attention module (EAM). Initial interest points are learned from the input features by the Interest Generation Module, while original offsets are generated by the Offset Generation Module from queries and refined by the Offset Guidance Module. Finally, sampled features are obtained by bilinear interpolation for keys and values.

Interest Generation Module

- Goal: Generate **differentiable interest points** to replace fixed reference points in deformable transformers.
- Lightweight module Net_{in} maps feature map F into coordinates (Px, Py) :

$$(Px, Py) = Tanh(Net_{in}(F))$$

- Output: $2 \times H_{in} \times W_{in}$ feature map
- Two channels = *horizontal* (Px) and *vertical* (Py) coordinates
- Kernel size k_{in} controls the number of interest points
- All coordinates are **normalized to** $(-1, +1)$ for training efficiency

Offsets Generation Module

- Computes offset values (Ox, Oy) for each interest point:

$$Q = F * W_Q$$

$$(Ox, Oy) = Tanh(Net_{of}(Q))$$

- Ensures **same spatial resolution** as interest map (matched kernel size $k_{of} = k_{in}$)
- Offsets help refine attention location flexibly

Offset Guidance Module

- Designed to **balance attention** between foreground and background
- First: Determine if **offset direction and interest point are in the same quadrant**:

$$w = 1 \text{ if } ReLU(OxPx) * ReLU(OyPy) > 0$$

$$\eta \text{ otherwise } (0 < \eta < 1)$$

- This **weakens attention drift** toward saliency if not aligned, encouraging background awareness

Offset Random Rotation (Early Training)

- Introduces randomness to **avoid local optima**:
- Randomly rotate offsets during early training epochs:

$$(O\bar{x}, O\bar{y}) = [w(Ox \cdot \cos\alpha - Oy \cdot \sin\alpha), w(Oy \cdot \cos\alpha + Ox \cdot \sin\alpha)]$$

$$\alpha = (1 - e/E) \cdot \text{Random}(-\pi, \pi)$$

- Enables **exploration of aesthetic cues** in background early on, stabilizes over time

Feature Sampling and Attention

- Final sampling coordinates:

$$(Px + O\bar{x}, Py + O\bar{y})$$

- Use **bilinear interpolation** to sample features:

$$F = B[F, \text{Tanh}(P + \bar{O})]$$

- Project sampled features:

$$K = \bar{F} * W_K V = \bar{F} * W_V$$

- Use standard multi-head attention on Q, K, V to compute final output

Overall EAT Architecture

- **Stage 1-2:** Swin Transformer blocks with shift window attention (local focus)
- **Stage 3-4:** EAT-enhanced blocks (global modeling between foreground and background)
- **Output head:** Predicts final aesthetic score

The architecture gradually shifts from **local aggregation** to **global relational modeling**

Design Rationale

- Avoid using EAT in early stages:
- May lead to **information loss** if applied before meaningful features emerge
- EAT blocks later in network:
 - Emphasize **foreground-background relations**
 - Preserve **background aesthetics** alongside salient features

Visualization Example

- Interest points visualized across training epochs
- **Circle size** represents attention weight
- Larger circles = higher attention by the model

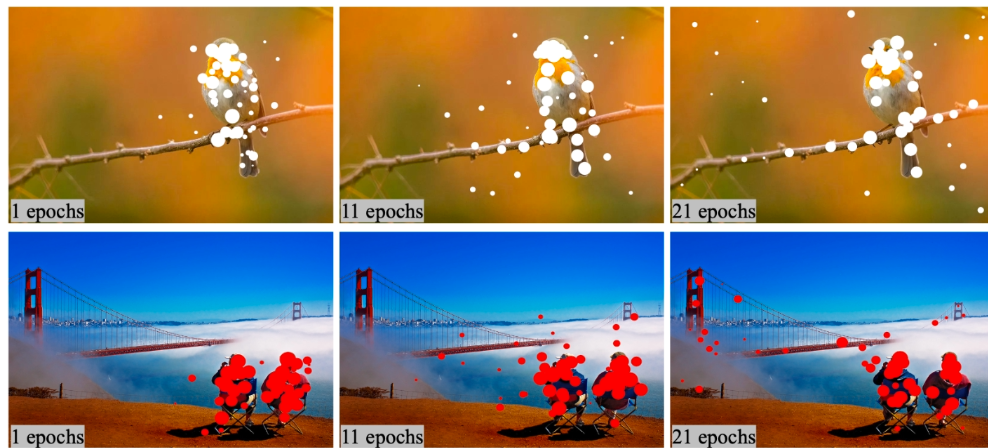


Figure 7: Interest points are visualized during training, with circle size indicating attention weights across multiple heads. Larger circles represent higher weights.

Visualization Example for Different Models

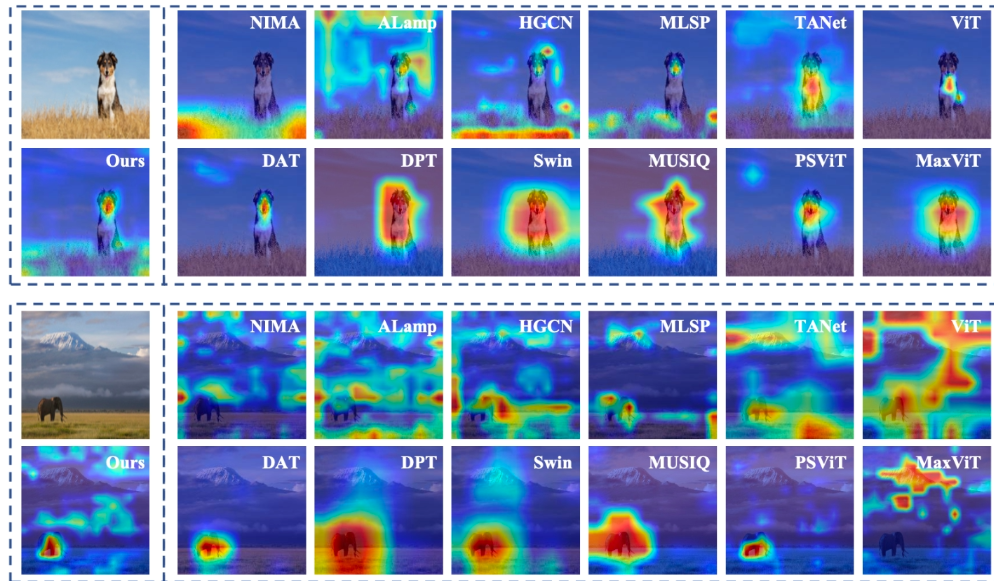


Figure 3: The saliency maps generated by 13 models for IAA tasks, while our EAT captures more reasonable and aesthetically relevant regions in both foreground and background by varying degrees of attention.

Experiment Result: Model Comparison on AVA

- Transformer-based models marked with * were re-trained for optimal performance. All models are pre-trained on ImageNet-1K. EAT was tested on 3 input sizes: 224×224, 512×512 (🟡), and 640×640 (🟢).

Metric (AVA)	CNN-based models								Tranformer-based models								Ours 224	Ours 512	Ours 640
	[34]	[26]	[2]	[30]	[11]	[10]	[13]	[12]	[38]	[3]	[21]	[15]	[15]	[37]	[35]				
	NIMA	ALamp	AFDC	HGCN	MLSP	TANet	ResNext	POC	PSViT*	DPT*	Swin	MUSIQ	ViT	DAT*	MaxViT				
	224	224	230	224	629	224	512	640	B224	L224	B224	512	B224	B224	512				
FLOPs	-	-	-	-	-	-	-	-	264G	264G	240G	111G	880G	240G	120G	140G	200G	280G	
Params	56M	99M	45M	44M	24M	40M	43M	1853M	21M	61M	87M	27M	88M	87M	31M	87M	87M	87M	
Epoch	40+	50+	-	60+	40+	110+	-	-	70+	100+	70+	90+	80+	60+	300+	25+	25+	30+	
PLCC↑	.636	.671	.671	.687	.757	.765	.781	.795	.645	.720	.737	.738	.739	.739	.745	.770	.790	.814	
SRCC↑	.612	.666	.649	.665	.756	.758	.780	.794	.701	.694	.736	.726	.728	.738	.708	.759	.786	.803	
Ratio↑	.751	.807	.779	.786	.925	.940	.940	.955	.847	.874	.901	.891	.913	.909	.877	.929	.950	.961	

Experiment Result: Model Comparison on TAD66K

- All models are pre-trained on ImageNet-1K.

Metric (TAD66K)	CNN-based models						Transformer-based models							
	NIMA 224	ALamp 224	AFDC 230	HGCN 224	MLSP 629	TANet 224	ViT* B224	PSViT* B224	DPT* L224	Swin* B224	MUSIQ* 512	MaxViT* 512	DAT* B224	Ours 224
Epochs↓	80+	150+	-	80+	76+	110+	120+	160+	160+	100+	90+	100+	80+	30+
PLCC↑	.405	.422	-	.493	.508	.531	.421	.431	.445	.454	.517	.518	.527	.546
SRCC↑	.390	.411	-	.486	.490	.513	.393	.408	70	.419	.489	.490	.499	.517
MSE↓	.021	.019	-	.020	.019	.016	.201	.022	.021	.201	.018	.018	.016	.015

Thank you! 🙏